Curriculum Vitae

YUHAO JIANG

Postdoctoral Researcher, EPFL

CONTACT

EPFL STI IGM RRL, MED 1-1017
Station 9 CH-1015 Lausanne, Switzerland
Phone: +41 79 350 5266
Email: yuhao.jiang@epfl.ch
Website: yuhaoj.com, Google Scholar, LinkedIn

RESEARCH INTERESTS

My research focuses on reconfigurable modular robotic systems as a pathway toward physical AI. I develop robots that adapt their physical embodiment to changing tasks and environments by integrating mechanical intelligence and embodied intelligence. On the hardware side, I study the co-design of mechanisms, materials, actuation, transmission, and embedded sensing to create compliant, multistable, and variable-stiffness systems capable of versatile behavior with fewer actuators. On the software side, I develop methods for multi-module embodiment planning, treating robot morphology and configuration as real-time planning variables optimized according to task goals, environmental conditions, and physical constraints. My long-term goal is to bridge digital AI and the physical world by developing novel robots as embodied interfaces through which intelligent systems can sense, understand, and act in real-world environments, contributing toward embodied artificial general intelligence.

EDUCATION

Arizona State University, Tempe, USA *Jan. 2019 – Aug. 2023*
Ph.D. in Mechanical Engineering
Advisor: Prof. Daniel Aukes
Dissertation: Design and Modeling of Soft Curved Reconfigurable Anisotropic Mechanisms

University of Florida, Gainesville, USA *Sep. 2015 – May 2017*
Master of Science in Mechanical Engineering

Donghua University, Shanghai, China *Sep. 2011 – Jun. 2015*
Bachelor of Engineering in Mechanical Engineering

PROFESSIONAL EXPERIENCE

EPFL, Lausanne, Switzerland *Sep. 2023 – Present*
Postdoctoral Researcher, Reconfigurable Robotics Lab
Supervisor: Prof. Jamie Paik

PUBLICATIONS

Journal Publications

[1] **Y. Jiang**, F. Chen, J. Paik, and D. M. Aukes, "Vibration of Soft, Twisted Beams for Under-Actuated Quadrupedal Locomotion," IEEE/ASME Transactions on Mechatronics, 10.1109/TMECH.2026.3686693.

[2] X. Zheng, **Y. Jiang**, M. Mete, J. Li, I. Watanabe, T. Yamada, and J. Paik, "Metamaterial Robotics," Science Robotics, 10, eadx1519 (2025). 10.1126/scirobotics.adx1519.

[3] **Y. Jiang**, S. Asmar, Z. Wang, S. Demirtas, and J. Paik, "CPG-based Manipulation with Multi-Module Origami Robot Surface," IEEE Robotics and Automation Letters, March 2025, 10.1109/LRA.2025.3555381.

[4] **Y. Jiang**, F. Chen and D. M. Aukes, "Tunable Dynamic Walking via Soft Twisted Beam Vibration," IEEE Robotics and Automation Letters, vol. 8, no. 4, pp. 1967–1974, April 2023, 10.1109/LRA.2023.3244716.

[5] M. Sharifzadeh, **Y. Jiang**, A. Lafmejani, K. Nichols, and D. M. Aukes, "Maneuverable gait selection for a novel fish-inspired robot using a CMA-ES-assisted workflow," Bioinspiration & Biomimetics, vol. 16, no. 5, pp. 056017, August 2021, 10.1088/1748-3190/ac165d.

[6] M. Sharifzadeh, **Y. Jiang**, and D. M. Aukes, "Reconfigurable Curved Beams for Selectable Swimming Gaits in an Underwater Robot," IEEE Robotics and Automation Letters, vol. 6, no. 2, pp. 3437–3444, April 2021, 10.1109/LRA.2021.3063961.

Conference Publications

[1] **Y. Jiang**, M. Sharifzadeh, and D. M. Aukes, "Shape Change Propagation Through Soft Curved Materials for Dynamically-Tuned Paddling Robots," 2021 IEEE 4th International Conference on Soft Robotics (RoboSoft), 2021, pp. 230–237, 10.1109/RoboSoft51838.2021.9479208.

[2] **Y. Jiang**, M. Sharifzadeh, and D. M. Aukes, "Reconfigurable Soft Flexure Hinges via Pinched Tubes," 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020, pp. 8843–8850, 10.1109/IROS45743.2020.9341109.

[3] D. Bailey, R. Moreno, S. Demirtas, Z. Wang, **Y. Jiang**, J. Paik, K. Stoy, and A. Faíña. "Flexible and Foldable: Workspace Analysis and Object Manipulation Using a Soft, Interconnected, Origami-Inspired Actuator Array." 2026 IEEE International Conference on Robotics and Automation (ICRA), 2026, Accepted, arxiv.org/abs/2509.13998.

[4] P. Bupe, **Y. Jiang**, J. Lin, T. Nguyen, M. Han, D. Aukes, C. Harnett, "Embedded Optical Waveguide Sensors for Dynamic Behavior Monitoring in Twisted-Beam Structures," 2024 IEEE 7th International Conference on Soft Robotics (RoboSoft), San Diego, CA, USA, 2024, pp. 139–144, 10.1109/RoboSoft60065.2024.10521938.

[5] M. Sharifzadeh, **Y. Jiang**, A. Lafmejani, D. M. Aukes, "Compensating for Material Deformation in Foldable Robots via Deep Learning – A Case Study," 2022 IEEE International Conference on Robotics and Automation (ICRA), 2022, 10.1109/ICRA46639.2022.9811752.

[6] M. Sharifzadeh, **Y. Jiang**, R. Khodambashi, D. M. Aukes, "Increasing the Life Span of Foldable Manipulators With Fabric." Proceedings of the ASME 2020 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. Volume 10: 44th Mechanisms and Robotics Conference (MR). Virtual, Online. August 17–19, 2020. V010T10A087. ASME, 10.1115/DETC2020-22757.

PATENTS

[1] "Pinched tubes for reconfigurable robots", D. Aukes, M. Sharifzadeh, **Y. Jiang**, N. Gravish, M. Jiang - US Patent US12017347B2

[2] "Buckling beams for underwater and terrestrial autonomous vehicles", D. Aukes, M. Sharifzadeh, **Y. Jiang** - US Patent US12240283B2

[3] "Mechanisms for steering robotic fish", D. Aukes, M. Sharifzadeh, K. Nichols, **Y. Jiang** - US Patent US11124281B2

[4] "Locomotion via vibration of soft, twisted beams with an under-actuated quadruped", D. Aukes, **Y. Jiang**, F Chen - US Patent Application US20260115895A1

[5] "Tunable Motion Using Flexible Twisted Beams", D. Aukes, **Y. Jiang**, F. Chen - US Patent Application 20240391542

SELECTED PROJECTS

MOZART: Morphing Computerized Mats with Embodied Sensing and Artificial Intelligence *Sep. 2023 – Present*
EPFL, Reconfigurable Robotics Lab
Funded by: EU Horizon Europe Research and Innovation Programme. Grant number: 101069536.

SCRAM: Soft Curved Reconfigurable Anisotropic Mechanisms *Jan. 2020 – May 2023*
Arizona State University, IdeaLab
Funded by: NSF EFRI C3 SoRo. Grant number: 1935324.

FIRE: Fish-Inspired Robots for Extreme Environments *Jan. 2019 – Jan. 2020*
Arizona State University, IdeaLab
Funded by: Salt River Project

GRANT ACTIVITIES

MusicTouch: Haptic Robotic Platform for Musical Experience
PI | EPFL–UNIL CROSS 2027 | CHF 60,000 | Awarded *Sep. 2026 – Sep. 2027*

COBRAS: Circular On-site Building with Robotic Autonomous Swarms
Co-applicant | EU EIC Pathfinder | EUR 4 million | Awarded *Nov. 2026 – Nov. 2029*

TEACHING AND STUDENT MENTORING

Course Instruction

ME410: Mechanical Engineering Product Design and Development *Fall 2023–2025*
Mechanical Engineering, EPFL

ME420: Advanced Design for a Sustainable Future *Fall 2024–2025*
Mechanical Engineering, EPFL

Graduate Student Mentoring

Gonçalo Pais *M.Sc. Mechanical Engineering | Master's semester project* *Spring 2025*
Development of Energy Storage and Release Mechanisms for Rapid Dynamic Motions in Canfield Origami Robots

Aurora Ruggeri *M.Sc. Mechanical Engineering | Master's semester project* *Spring 2024*
Study of soft metamaterials for object sensing and geometry generation

Louis Flahault *M.Sc. Robotics | Master's semester project* *Spring 2024*
Kinematic study and design of a spatially reconfigurable modular robotic platform

Serge Asmar *M.Sc. Robotics | Master's semester project* *Spring 2024*
Locomotion design and control using surface-wave changes generated by the Ori-Pixel platform

Nicolas Nouel *M.Sc. Robotics | Master's thesis | Co-advisor* *Spring 2024*
Programmable surface using bistable structures

TALKS

Conference Proceedings Talks

[1] **IEEE-IROS 2025:** "CPG-based Manipulation with Multi-Module Origami Robot Surface"

[2] **IEEE-RoboSoft 2023:** "Tunable Dynamic Walking via Soft Twisted Beam Vibration"

[3] **IEEE-ICRA 2022:** "Compensating for Material Deformation in Foldable Robots Via Deep Learning – a Case Study", https://youtu.be/AwS4vabv-JQ

[4] **IEEE-ICRA 2021:** "Reconfigurable Curved Beams for Selectable Swimming Gaits in an Underwater Robot", https://youtu.be/EszTDc9slyw

[5] **IEEE-RoboSoft 2021:** "Shape Change Propagation Through Soft Curved Materials for Dynamically-Tuned Paddling Robots"

[6] **IEEE-IROS 2020:** "Reconfigurable Soft Flexure Hinges via Pinched Tubes", https://youtu.be/J5heXXD6mVo

Workshop Presentations

[1] **IEEE-RoboSoft 2025:** "CPG-Based Motion Generator for Multi-Module Origami Robot"

[2] **IEEE-RoboSoft 2023:** "Model Order Reduction for Vibrational Soft Twisted Beams Using Pseudo-rigid-body Modeling – A Case Study" (**awarded 2nd place for best talk**)

[3] **IEEE-ICRA 2022:** "Modular Robots Using Soft Curved Reconfigurable Anisotropic Mechanisms"

Seminar Talks

[1] "Empowering Actuation of Soft Robotic Systems via Soft Curved Reconfigurable Anisotropic Mechanism", hosted by Prof. Nick Gravish and Prof. Michael Tolley, UCSD, Feb. 2023.

ACADEMIC SERVICE

Event Organization

[1] **IEEE-IROS 2025 Workshop :** "S'MORE: Shape-Morphing Robotics via Embodied Sensing and Mechanisms", https://www.shapemorphingrobot.com/

[2] **2024 RRL Demo Day:** Full-day public event for research projects from RRL and ME-410 and ME-420 class,
https://www.paikslab.com/courses

[3] **2023 RRL Demo Day:** Full-day public event for research projects from RRL and ME-410 class, https://www.paikslab.com/courses

[4] **Robosoft 2021 Workshop :** "Breaking the Mold: Challenging Current Paradigms in Soft Robotics", https://www.scrambots.com/robosoft-2021-workshop

Media Interview

[1] **RTS Education and Scientific Program:** feature in "A guide to the future: Swiss Federal Institute of Technology 02", https://www.youtube.com/watch?v=9yoNLg5Qho0&t=560s

Editor

IEEE-IROS 2026: Associate Editor

Journal Reviewer

The International Journal of Robotics Research (IJRR)
IEEE Transactions on Robotics (T-RO)
IEEE/ASME Transactions on Mechatronics (T-Mech)
IEEE Robotics and Automation Letters (RA-L)
Soft Robotics (SoRo)
Journal of Field Robotics (JFR)
ASME Journal of Mechanisms and Robotics (JMR)

Conference Reviewer

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
International Conference on Robotics and Automation (ICRA)
International Conference on Soft Robotics (Robosoft)
ACM Symposium on Computational Fabrication (SCF)

Grant Reviewer

Swiss National AI Institute (Seed Grant Program)

Demos and Expositions

[1] RRL lab tours ($\sim$ 6 times per year)

[2] Swiss Robotics Day (2023–2025)